DTU

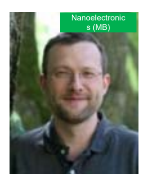
Feb	Mar	Apr	Maj
L 1	S 1	O 1	SYNOPSIS
S 2	Electron-phonon	Frameworks (PB/MB)	L 2
Introduction PB/TB	T 3	F 3	S 3
T 4	O 4	L 4	Innovation (PB)
O 5	Nanoelectronics (MB)	S 5	T 5
Synthesis and statbility PB/AS	F 6	M 6	O 6
F 7	L 7	T 7	Feedback
L 8	S 8	O 8	F 8
S 9	Nanoelectronics (MB)	T 9	L 9
Carbon sp2 chem. KA	T 10	F 10	S 10
T 11	O 11	L 11	Innovation (PB)
O 12	Quantum tramsport (APJ)	S 12	T 12
Defects and GFET (PB)	F 13	M 13	O 13
F 14	L 14	T 14	HANDIN
L 15	S 15	O 15	F 15
S 16	vdW heterostructure	2D Photonics (NS)	L 16
Quantum tramsport (APJ)	T 17	F 17	S 17
T 18	O 18	L 18	M 18
O 19	Advanced devices	S 19	EXAM
Quantum tramsport (APJ)	F 20	2D Photonics (NS)	O 20
F 21	L 21	T 21	T 21
L 22	S 22	O 22	F 22
S 23	JCRPW2	2D Photonics (NS)	L 23
JC/RPW1	T 24	F 24	S 24
T 25	O 25	L 25	M 25
O 26	Advanced devices	S 26	T 26
Quantum tramsport (APJ)	F 27	2D Photonics (NS)	O 27
F 28	L 28	T 28	T 28
L 29	S 29	O 29	F 29
	2D material discovery (KST)	JCRPW3	L 30
	T 31		S 31

Prof. middag

Teachers

PB Peter Bøggild
DTU Fysik

APJ Antti Pekka Jauho
DTU Fysik



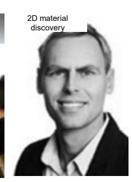
MB Mads Brandbyge
DTU Fysik

NS Nicolas Stenger
DTU Fotonik

Guest teachers

Tim Booth
DTU Fysik

Kristoffer Almdal
DTU Chemistry



Kristian Thygesen
DTU Fysik

Aravind Vijyaraghavan
Univ. Manchester

Introduction

Nano 3

Journal club and research proposal workshop

Part 1: idea generation
Journcal club (3 presenters)
Part 2: research proposal and assignment

Schedule

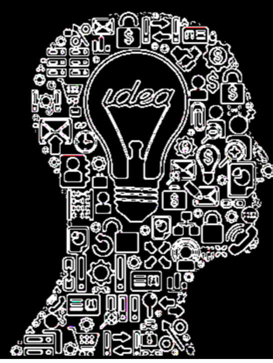
13.00 Introduction research proposals workshop (RPW) part 1: idea generation
13.30 Journal club (13.30 Peder, 14.10 Cheng, 14.50 Martin)

13.30	**25 minutes (Peder)**	**Presentation 1**	**Cover all 4 levels (descriptive, comparative, evaluative, creative)**
	10 minutes (Christina)	Review and open discussion	
	5 minutes	Feedback and break	
14.10	**25 minutes (Cheng)**	**Presentation 2**	**Cover all 4 levels (descriptive, comparative, evaluative, creative)**
	10 minutes (Andreas)	Review and open discussion	
	5 minutes	Feedback and break	
14.50	**15 minutes (Martin)**	**Presentation 3**	**Cover 2 levels (descriptive, comparative)**
	10 minutes (Simone)	Questions and open discussion	

15.30 Discussion of papers
16.00 Research proposals workshop (RPW) part 2 : the assignment
17.00 Home!

The idea

13.00-13.15



Discuss

- How do you get ideas? When?
- What makes an idea a ***good*** idea?
- What do you do with your ideas?

How to get an idea - unstructured

- Relax - Comes when doing something else than work
- Study - others work and proposals
- Discuss – helps generate and shape new ideas
- Allow your mind and discussion to wander
- Generate <u>many</u> ideas, and sort them afterwards
- ”AHA moments” are rare – ideas are usually incremental

Sometimes the best ideas hide behind a lot of bad ideas

Kandel: "Preparation is the period where we consciously work on a problem," Kandel writes, "and incubation is the period when we refrain from conscious thought and allow our unconscious to work.“

Pauling: The best way to have a good idea is to have lots of ideas.

Mozart: When I am, as it were, completely in myself, entirely alone, and of good cheer -- say traveling in a carriage, or walking after a good meal, or during the night when I cannot sleep; it is on such occasions that my ideas flow best and most abundantly. Whence and how they come, I know not; nor can I force them. All this fires my soul, and, provided I am not disturbed, my subject enlarges itself, becomes methodized and defined...All this inventing, this producing, takes place in a pleasing lively dream.

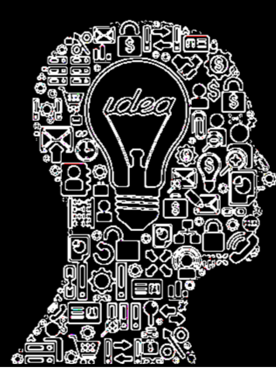
Myths about creativity

Myth #1: Creative individuals are born, not made
Myth #2: Innovation is the sole purview of the lone creative genius
Myth #3: Intellectual work is not creative work
Myth #4: Creative work is totally original
Myth #5: Creativity is open and free and cannot
thrive in contexts of constraint
Myth #6: Creative work is not practical

http://www.utexas.edu/ogs/publications/toolkit_vol2.pdf

Generation

13.15-13.30



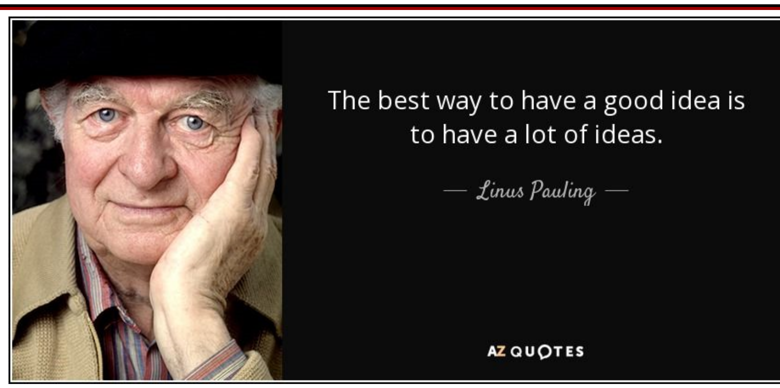
Ideas build on other ideas

“No genius invented their parents. No legendary artist created the idea of art or language. Mozart didn’t invent the piano and Picasso didn’t invent paint. We have always borrowed, reused, and remade, standing on the shoulders of ideas that came before us and we always will.

It’s impossible to find any idea that can’t be broken down into smaller ideas. Pick a song, an invention, a philosophy… they are all recombinations of other ideas. Sometimes the grandest ideas, once you ditch the romance, are the easiest to dissect. Until you can see the fact that all ideas are made of other ideas as an immutable law, you’re unlikely to create, since you see ideas in the world as fixed when in truth they are always in motion.”

Scott Berkun, author.

Good ideas build on many bad ideas



The subjects of the papers he published reflect his great scientific versatility: about 350 publications in the fields of experimental determination of the structure of crystals by the diffraction of X-rays and the interpretation of these structures in terms of the radii and other properties of atoms; the application of quantum mechanics to physical and chemical problems, including dielectric constants, X-ray doublets, momentum distribution of electrons in atoms, rotational motion of molecules in crystals, Van der Waals forces, etc.; the structure of metals and intermetallic compounds, the theory of ferromagnetism; the nature of the chemical bond, including the resonance phenomenon in chemistry; the experimental determination of the structure of gas molecules by the diffraction of electrons; the structure of proteins; the structure of antibodies and the nature of serological reactions; etc. (goes on and on)

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SCAMPERR by Alex Osborn

Substitute: Remove some part of the accepted situation, thing, or concept and replace it with something else.
- *What if doctors are paid based on performance*

Combine: Join, affiliate, or force together two or more elements of your subject matter and consider ways that such a combination might move you toward a solution.
- *Combining watch with smart phones smart watch*
- *Islamic religious leaders mobilized for family planning*

Adapt/adjust: Change some part of your problem so that it works where it did not before.
- *Telemedicine where specialists are lacking,*
- *Change clinic start times , Client reminder letters/calls*

Modify: Consider many of the attribute of the thing you're working on and change them, arbitrarily, if necessary. Attributes include: size, shape, other dimensions, texture, color, attitude, position, history, and so on.
- *What if we change premium, what if we change benefit package*

Put to other use: Modify the intention of the subject. Think about why it exists, what it is used for, what it's supposed to do. Challenge all of these assumptions and suggest new and unusual purposes.
- *Can certain types of services be delegated to nurses*

Eliminate: Arbitrarily remove any or all elements of your subject, simplify, reduce to core functionality
- *What if we eliminate users fee in health facilities*
- *Reduce information gathered on repeat client health history forms*

Reverse/Rearrange: Change the direction or orientation. Turn it upside-down, inside-out, or make it go backwards, against the direction it was intended to go or be used.
- *Require late patients to reschedule rather than be worked into the schedule.*

https://www.slideshare.net/achyutrajpandey/research-idea-generation

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Steps of generating research ideas

- Do some background reading
- Start narrowing down your area
- Mapping ideas
- Refining the ideas
- Researching the main ideas through literature review
- Taking notes and deciding the topic
- Examining originality

Translating idea to research

- For the idea to develop; research idea should be of genuine interest to researcher.
- *"its not possible to have a research question in mind and not be curious at the same time"*
- After you come up with an idea, you need to decide on what aspects of that idea you will emphasize.

https://www.slideshare.net/achyutrajpandey/research-idea-generation

Idea games

Games/methods to come up with research ideas for 2D Materials…

1. **Mutation game:** read an article, and see if you can make it more exciting by changing method, material or application?
2. **Profession game: t**hink about one profession, and a tool, material or machine used by this profession. Can a 2D material be used to improve it?
3. **”Super” game:** Think about one superior property of graphene/2D materials – list everything you can think about where this property would be useful. Dont be critical.

Pro tip: Write everything down… sort out your catch later…

Now comes the **sorting game: novelty, impact, feasibility – and your interest!**

Mutation game

– Can the original idea be changed, by another material, another application, another method
– Could your special competences be used?
– Could the facilities at your work be used?
– Can you combine it with another idea?
– Can you think of a way to mass-produce it?
– Or use it for something other than the originators thought of?

Everyone except presenter and reviewer:

- Find at least one mutation of each paper → new idea…

Your ideas

15.30-16.00

The proposal

16.00-16.20

Write proposal – because you will (probably) need to in your job

- Your job is often to **solve a difficult problem** or answer a difficult question
- Maybe you have a **great idea** on how – or need to get one!
- If you do not have money or time, you need to **convince** someone who has
- A proposal is a written document, where you **ask for ressources** to try out your idea/product/solution

Research proposals – why?

Examples

1. **Travel money**
2. **Equipment**
3. **Postdoc stipend at national agency**
4. **Marie Curie mobility stipend – postdoc abroad?**
5. **Proposal for external stay at a relevant laboratory**
6. **A PhD project – research plan**
7. **An internal proposal in a company**
8. **Large European project (with many other partners)**
9. **A pitch to investors or patent office**

… many many more

What you need before you start

Important:

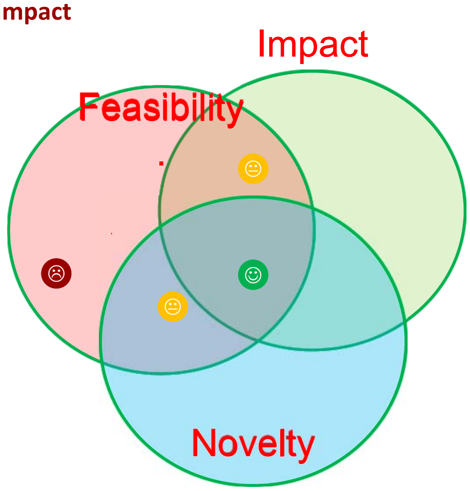
- IMPACT - A an idea (goal or approach) which is important
- NOVELTY – The idea or approach must have some originality
- FEASIBILITY - A plan and ressources that makes it realistic

But also:

- FIT the FUNDING SOURCE (more about this later)
- DO ALL FORMALITIES TO THE LETTER

Novelty, Truth/Feasibility, Impact

"It can be done"
"I can do it"
"I have excellent ressources"
"I manage risks"
"I have unique opportunity, because…"



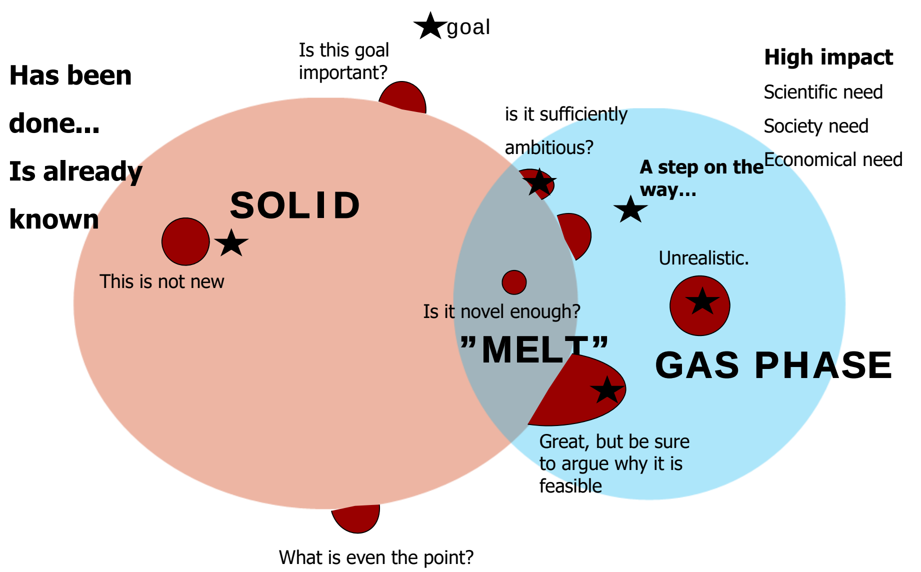
"This will enable us to…"
"This will teach us a lot…"
"This has impact on society/life/science"
"This will answer a big question…"

"Nobody tried it this way"
"They tried but failed"
"Nobody thought of this"
"Its completely new"

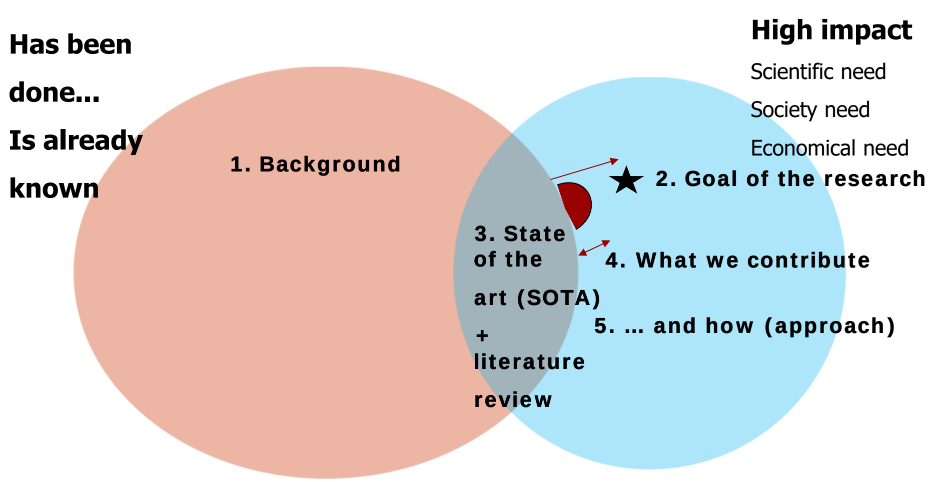
Convince us that the idea is

- Relevant — and will have high impact –on society, on science, and you
- Novel — in some important way, compared to state of the art
- Real(istic) — due to your skills, your facilities, its uniqueness, your collaborators – but still ambitious

Phase diagram of scientific knowledge



to the S.O.T.A.



Research proposals – to evolve

A proposal is a way to mature an idea and transform it into a plan

It is a manifest, and can help you to define your profile

Your next employer will see it as an attractive skill

Investment - A good proposal usually gets funded sooner or later

Writing a proposal is a way to define your own job

A proactive way to apply for a job

Even in a business without formal procedures for screening and selecting internal ideas, writing good proposal is a powerful tool for you

Which other professions pay the ”worker” to specify what the work should be about?

Schedule

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15.30 Discussion of papers

16.00 Research proposals workshop (RPW) part 2 : the assignment

17.00 Home!

24-02-2020

The assignment

16.20-16.40

Issues

You should be clear on where you and your proposal stand on

- Idealism vs pragmatism
- Explorative vs Planned research
- Innovator vs Improver
- Follow the money vs follow your vision?

… and whether if aligns with the funding agency…

FUNDING AGENCY

It depends… (agency, topic, call)

1) What do they want?
2) Who do you know who has tried submission or refereeing?
3) Can *your* idea be better aligned to *their* requirements?
   1) Products vs knowledge?
   2) Create a product, or your career are the product?
   3) Technology or knowledge (or both)

THE REVIEWER/EVALUATOR

The proposal <u>may</u> be read by someone, who do not
- have deep insight
- have enough time to read your proposal
- have enough money to fund all good deserving proposals
- like you or your project
- Care

…Or may be a **real expert**

You need to anticipate and overcome all this
- **Do your homework**
- **Be clear and concise**
- **Write important things first**

HOW TO MAKE A SUCCESFUL WORKPLAN

- **Constraints: read first any information on**
  - Inquire about funding agency – what kind of proposals do they grant
  - Who has succesfully applied – talk to them, ask them what they did
  - Scope / topics – which areas can be funded, can your idea be fitted
  - Formal requirements – proposal layout/length, your status/age etc.
  - Deadlines – make a realistic plan, plan long time ahead and start early

- **Set the team (colleagues, collaborators, supervisor)**
  - Make sure everybody knows what to do and has the time for it
  - Make a roadmap and have it approved
  - Get support letters and support text in early

Change their mind (”mind manipulation”)

If you succeed in getting funding, you have convinced the agency/reviewers that

- The idea was NOVEL, RELEVANT and CREDIBLE
- It NEEDS to be done
- The time is NOW
- YOU are the only one who can do it
- If they dont act, it is a HUGE MISSED OPPORTUNITY, and THEY will be responsible
- You are on the same team!

Typical proposal

Typical elements:

- Resume/abstract – short description of the proposal
- Introduction – the field, state of the art and THE GAP
- Hypothesis / question
- Goals
- Scientific background
- Technical foundation
- Methodology
- Plan and risks
- Impact(s): on you, research field, society, industry?
- Applicant, Facilities and Environment
- Budget
- Letters of reference

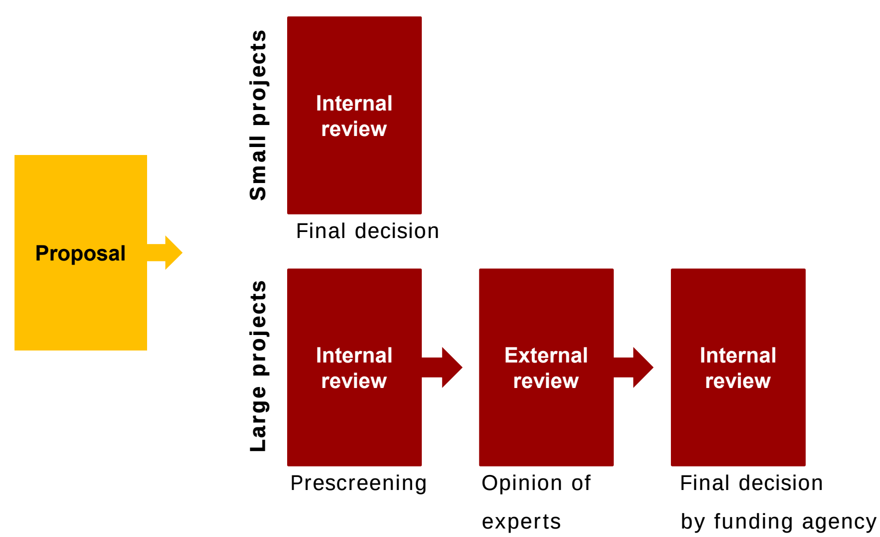
NANO 3 (technical proposal)

In this assignment

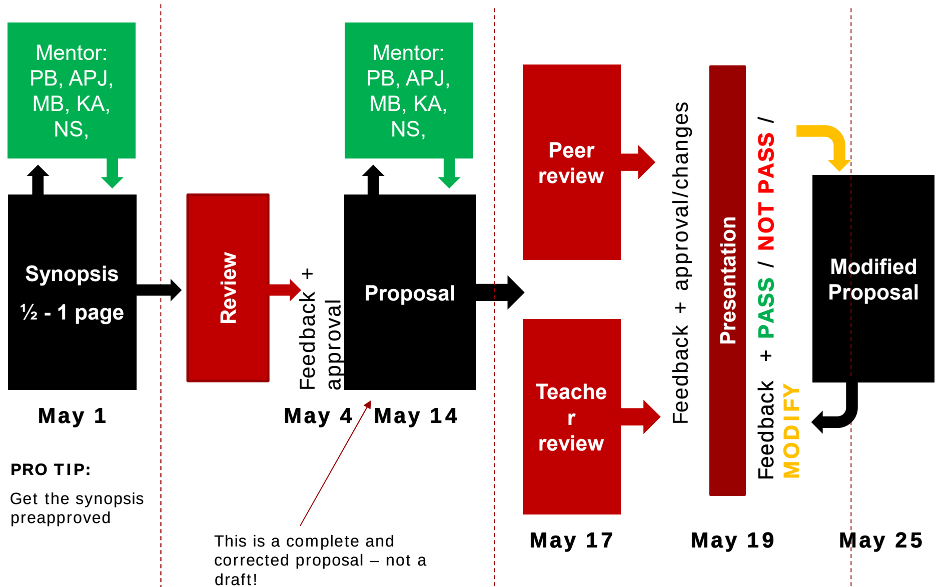
- Resume/abstract – short description of the proposal
- Introduction – the field, state of the art and the ”leap of novelty”
- Hypothesis / question
- Goals
- Scientific background
- Technical foundation
- Methodology (focus point)
- Plan and risks
- Impact(s): on you, research field, society, industry?
- Applicant, Facilities and Environment
- Budget
- Letters of reference

The proposal	What the reviewer will look for
Resume/abstract – short description of the proposal	Is it brief, clear and powerful? Do I understand what the proposal is about?
Introduction – the field, state of the art and THE GAP	Is the field well described, as well as why it is important? Are the most important contributions mentioned? Is it clear what has not been done and why this is crucial happens?
Hypothesis / question	Is the research hypothesis clear, accurate and powerful? Do I even want to know the answer?
Goals	Are the goals SMART: specific, measurable, ambitious, realistic and timed? (they dont HAVE to be, though)… Are they relevant?
Methodology/Scientific-technical foundation	Which methods will the applicant apply? Are they relevant? Are there any flaws? Does the applicant know what he/she is talking about? Are the expected outcomes realistic?
Plan and risks	Is the plan clear, not overly detailed? Is it realistic? Are the milestones/phases relevant? Are the risks described, and how to deal with them? What are the plan B's, for the largest risks?
Impact(s): on you, research field, society, industry?	Are the projected impacts important? Is it overly optimistic or imaginary? Are there nice secondary impacts (education of students, outreach)
Applicant, Facilities and Environment	Is the candidate qualified? Should someone else rather do the project? Is this a team or an individual? Are the facilities there? Is there a strong network?
References	Is it under or overwhelming? Is it free of errors? Any omissions?

Research proposal – ”real” version



Research proposal – Nano 3 version



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15.30 Discussion of papers

16.00 Research proposals workshop (RPW) part 2 : the assignment

17.00 Home!

Preparation until next RPW session (march 23)

Read one of the uploaded research proposals

- Novelty: what is the new idea, and how is the novelty argued?
- Impact: what problem does it solve? Is it important to solve? Do you think the proposal makes a strong case for this?
- Realistic: how does the proposal argue that it can be done, and does that sound convincing?
- Mutation: what would you change to make it a better proposal?
- **USE review template**

Feedback